PROJECT SALUS

Counter-UAS Site Coverage Report

Scenario	**compound_terrain**
Generated	2026-04-10T00:23:53Z
Sensors deployed	4
Composite coverage	**47.5%**

47%
Composite Coverage

4
Sensors Deployed

This report was generated automatically by Project Salus. Results are based on modelled sensor capabilities and terrain data. See the Assumptions and Limitations section for constraints and caveats applicable to this analysis.
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Executive Summary

The assessed sensor configuration provides 47.5% composite coverage across the site, representing a limited detection capability. The total uncovered area is 1.182 km², with the largest contiguous gap measuring 117.94 ha. Significant coverage gaps exist. Additional sensors or a revised placement strategy are strongly recommended before operational deployment.

Coverage by sensor modality is as follows: EO_IR: 43.2%, RF: 13.3%, Radar: 4.7%. Combining multiple sensor modalities improves overall detection probability and reduces the risk of a single-modality gap being exploited by an adversary.

No named zone analysis was performed for this configuration. Defining named zones for critical assets (runways, buildings, perimeter boundary) enables per-asset coverage metrics and more targeted gap prioritisation. Zone analysis can be added by providing a GeoJSON zones file in the scenario.

47.5%	118 ha	118 ha
Total Coverage	Gap Area	Largest Gap

Site Overview — Composite Coverage Map

Composite Coverage

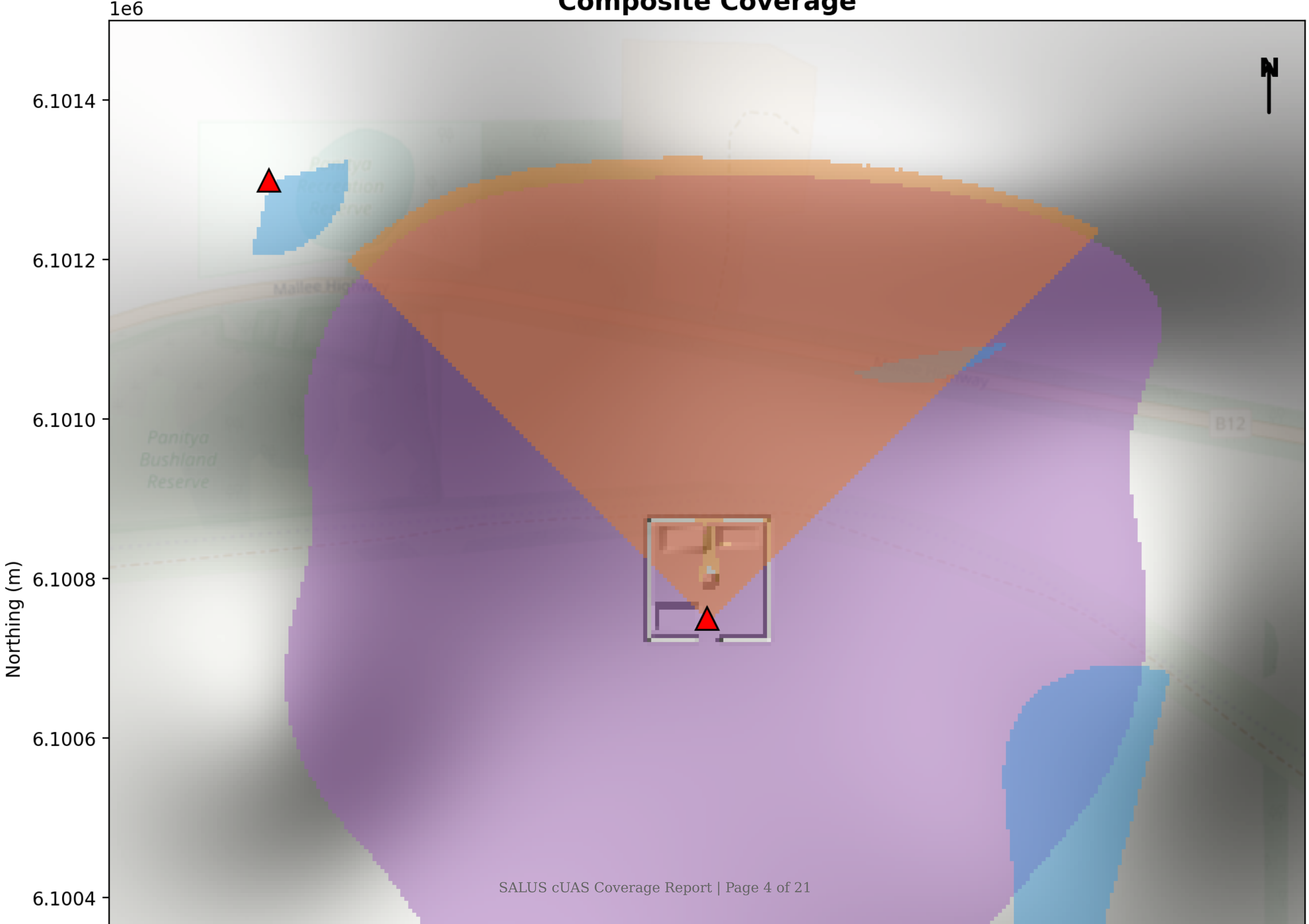


Figure 1. Composite sensor coverage map for scenario compound_terrain. Each sensor modality is shown in its designated colour; overlapping layers are additive.

Metric	Value
Total composite coverage	47.5%
Gap area	1,181,900 m²
Largest contiguous gap	1,179,425 m²
Sensors deployed	4
Sensor modalities	EO_IR, Radar, RF
Report generated	2026-04-10T00:23:53Z

Coverage Analysis

Per-Layer Coverage

Sensor Modality	Coverage (%)
EO_IR	43.2%
RF	13.3%
Radar	4.7%
Composite (any modality)	**47.5%**

Layer Coverage Maps

EO_IR Layer

EO_IR Layer Coverage

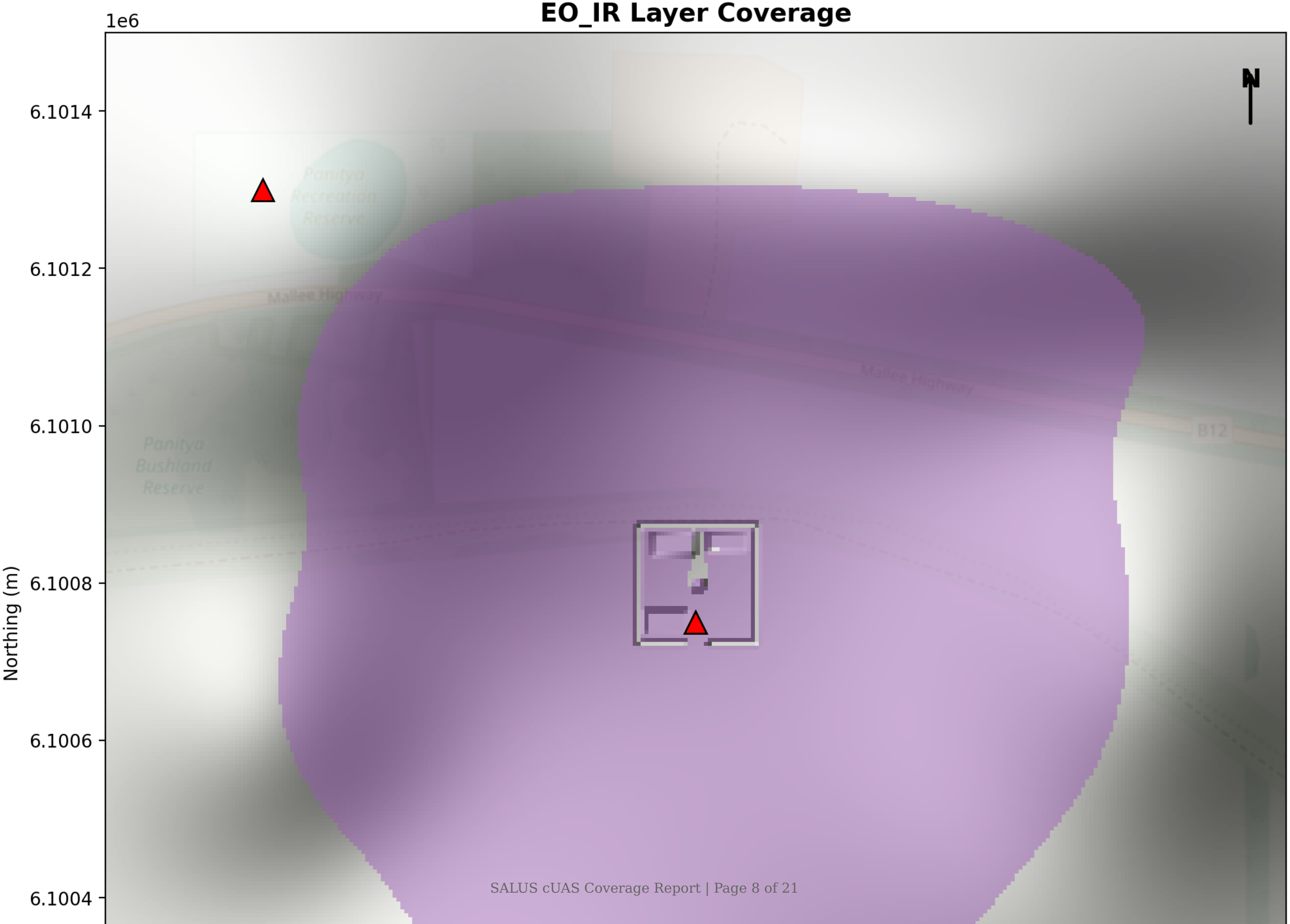


Figure. EO_IR sensor-type coverage layer.

Radar Layer

Radar Layer Coverage

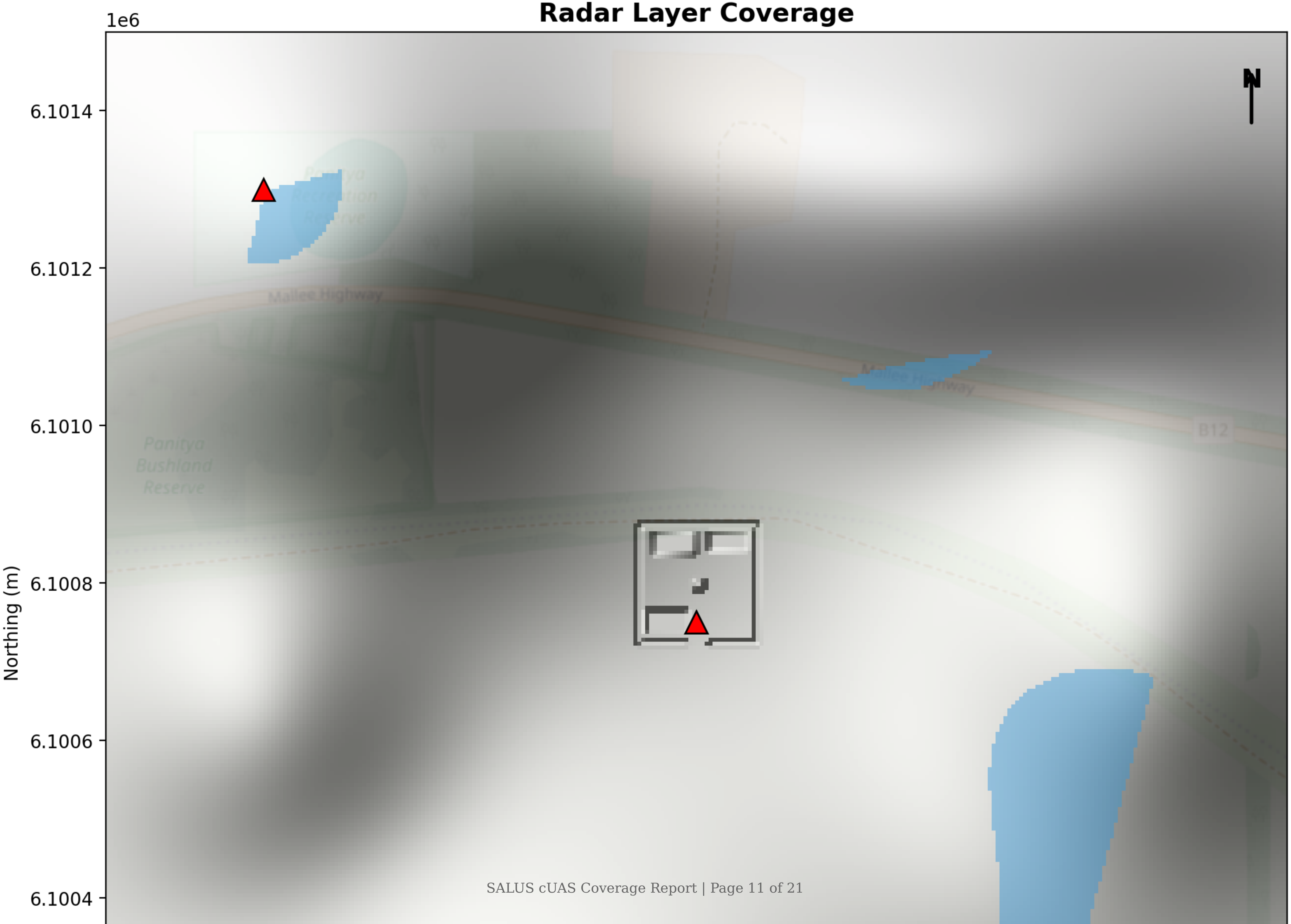
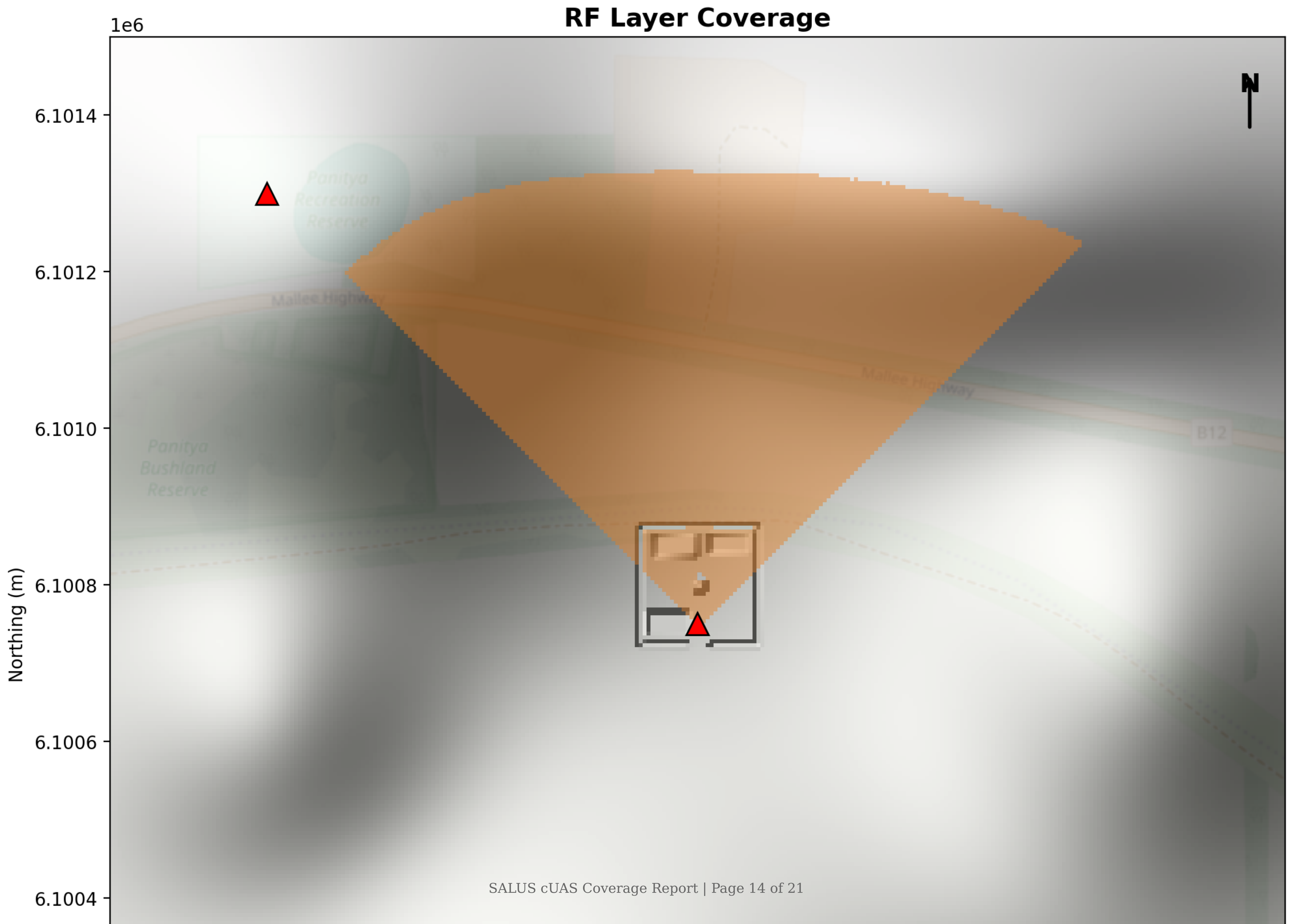


Figure. Radar sensor-type coverage layer.

RF Layer



RF Layer Coverage
1e6
6.1014
6.1012
6.1010
6.1008
6.1006
6.1004
Northing (m)
N
Panitya Recreation Reserve
Mallee Highway
Panitya Bushland Reserve
B12

Figure. RF sensor-type coverage layer.

Coverage Gap Analysis

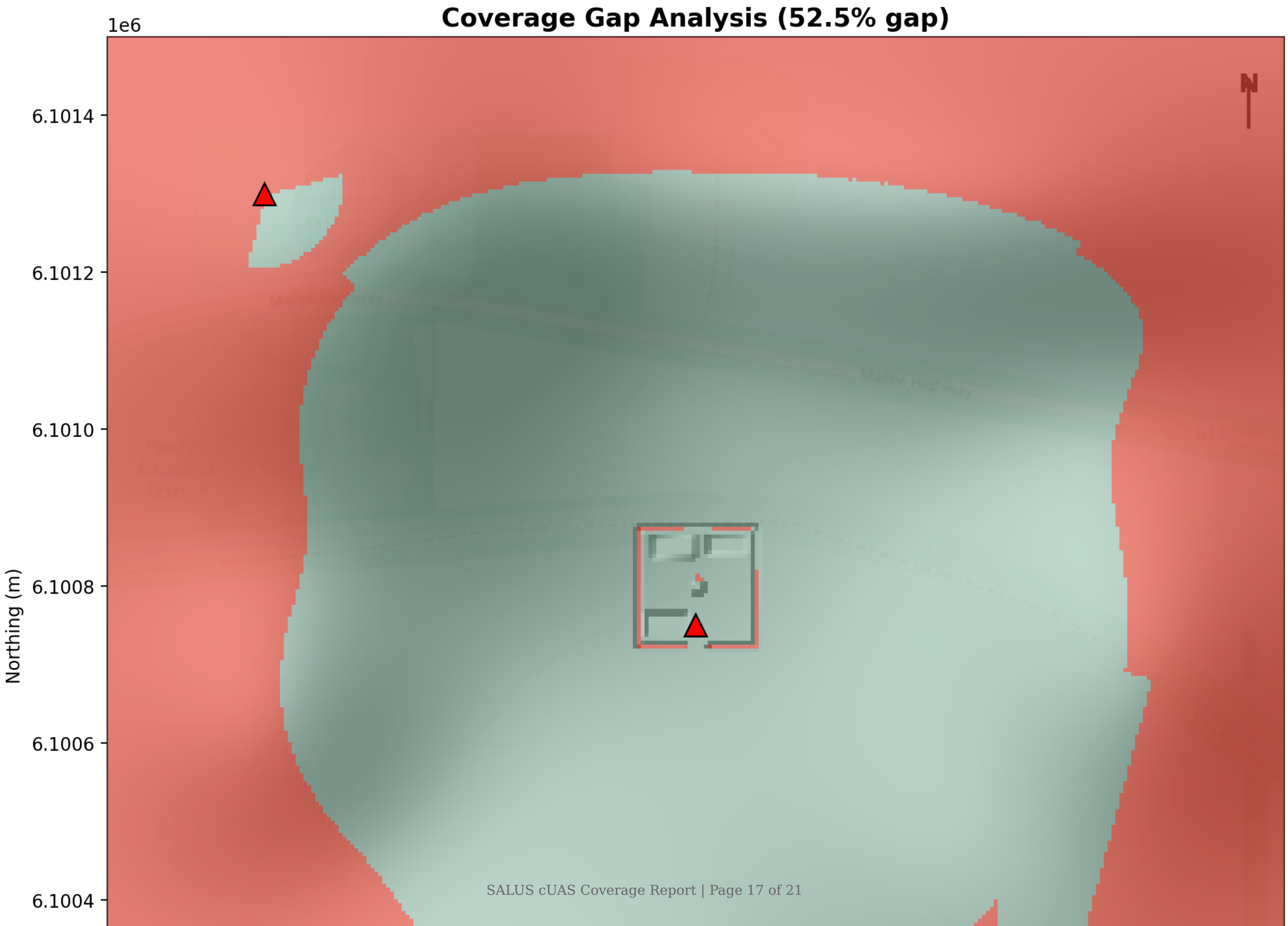

Coverage Gap Analysis (52.5% gap)
1e6
6.1014
6.1012
6.1010
6.1008
6.1006
6.1004
Northing (m)
N
Mallee Highway

Figure. Coverage gap map. Red areas are inside the site boundary but not detected by any sensor. Grey-green areas are covered by at least one sensor modality.

Gap Metric	Value
Total gap area	1,181,900 m² (52.5%)
Largest contiguous gap	1,179,425 m² (117.94 ha)

Assumptions and Limitations

This analysis is based on modelled sensor capabilities and terrain data. The following assumptions and limitations apply to all results presented in this report. Consumers of this report should review these caveats before drawing operational conclusions.

Terrain Model

- Terrain model: Digital Elevation Model (DEM) at 5.0 m/pixel resolution (EPSG:28354)
- No vegetation layer — canopy attenuation not modelled; results may be optimistic in forested areas

Sensor Modelling

- Sensor specifications sourced from manufacturer data sheets unless noted otherwise
- Detection model: binary (detected / not-detected) — no probability-of-detection (Pd) curves applied; detection is either certain or impossible at range boundaries
- All sensors modelled as stationary at their configured mounting height above ground
- Azimuth coverage is a hard arc boundary — beam-shape, side-lobe, and antenna pattern effects are not modelled
- Sensor elevation arc is a hard cone boundary — target elevation angle must fall within the configured arc; no tapering at arc edges

Propagation Model

- Free-space propagation model — multipath, diffraction, and atmospheric refraction are not modelled
- No atmospheric attenuation applied — rain, fog, temperature gradients, and humidity effects on RF and acoustic propagation are excluded
- RF propagation uses line-of-sight assessment only — ground-wave and ionospheric propagation paths are not considered
- Radar cross-section (RCS) assumed constant across all aspect angles — target attitude, material, and orientation are not modelled
- Acoustic propagation uses free-field spherical spreading — wind noise, turbulence, and ground-reflection effects are excluded

Threat Model

- Threat approach modelled as constant-altitude, constant-speed, straight-line flight
- No evasive manoeuvres, terrain-following, or nap-of-the-earth flight modelled
- Threat altitude taken from profile typical_altitude_m — actual operational altitude may differ
- Threat radar cross-section and RF emission profile sourced from published threat data; actual values may vary by operator modification
- Kill-chain timings are deterministic — operator reaction-time variability and comms latency are not modelled

General Caveats

- This report does not constitute an operational risk assessment and should not be used as the sole basis for procurement or deployment decisions.
- Results are valid only for the specific terrain model, sensor placement, and threat profiles specified in the scenario. Changes to any of these inputs require a new analysis.
- This analysis does not account for electromagnetic interference (EMI), jamming, spoofing, or electronic warfare effects on sensor performance.
- Sensor degradation due to weather, maintenance state, or operator fatigue is not modelled.

Appendix A — Sensor Specifications

The following table lists all sensor definitions used in this analysis. All values are as-configured in the sensor database. Fields marked † are manufacturer-supplied values; fields marked ‡ are conservative defaults used where vendor data was unavailable.

Sensor Name	Type	Max Range (m)	Az. Coverage (°)	El. Coverage (°)	Mount Ht (m)	LOS Required	Veg. Penetration
Anduril WISP	EO_IR	5,000	360	125	5.0	Yes	0.0
DroneShield DroneSentinel	Acoustic	300	360	90	2.0	No	0.9
DroneShield DroneSentry-X Mk2	RF	8,000	360	180	3.0	No	0.6
DroneShield RfOne Mk2	RF	8,000	90	60	5.0	No	0.6
Echodyne EchoGuard	Radar	900	120	80	5.0	Yes	0.4
HENSOLDT Spexer 500	Radar	3,000	120	65	5.0	Yes	0.2

Appendix C — Sensor Deployment

Sensor Name	Easting (m)	Northing (m)	Bearing (°)	Height Override (m)
Anduril WISP	500,750	6,100,750	0	22.0
Echodyne EchoGuard	500,200	6,101,300	135	5.0
Echodyne EchoGuard	501,200	6,100,200	315	5.0
DroneShield RfOne Mk2	500,750	6,100,750	0	8.0